

MATH 326 / MTHE 326 — Functions of a Complex Variable

Course Outline — Fall Term 2026

Department of Mathematics and Statistics, Queen's University

1. General Course Information

Item	Detail
Course title	Functions of a Complex Variable
Course code	MATH 326 / MTHE 326 (cross-listed)
Course components	Lecture (3 hours per week) Tutorial (1 hour per week)
Year and term	Fall Term 2026 (8 September – 8 December 2026)
Number of units	3.0 units (MATH 326) / 3.5 units (MTHE 326)
Mode of instruction	In person
Lecture times and location	Mondays 12:30 – 13:30 Thursdays 13:30 – 14:30 Fridays 10:30 – 11:30 Kingston Hall 201
Tutorial time and location	Tuesdays 9:30 – 10:30, Kingston Hall 201
Course website	onQ (Queen's learning management system)

2. Instructor Information

Item	Detail
Instructor	Ivan Dimitrov
Title	Professor, Department of Mathematics and Statistics
Email	dimitrov@queensu.ca
Office	Jeffery Hall, Room 508
Office hours	Mondays 11:00 – 12:00 and by appointment

Communication policy

Email is the best way to reach me outside of class. I aim to reply to course-related email within two business days. I do not normally read or answer email in the evenings, on weekends, or on university holidays. Questions about mathematics are usually answered far more efficiently in person, so please

bring them to office hours, to the tutorial, or to the end of a lecture. Course-wide announcements will be posted on onQ.

3. Course Description

Calendar description

Complex numbers, analytic functions, harmonic functions. Cauchy's theorem. Taylor and Laurent series. Calculus of residues. Rouché's theorem.

Prerequisite: MATH 281 / MTHE 281.

Expanded description

This is an introduction to complex analysis: the calculus of functions of one complex variable. The subject begins with a modest-looking change — allowing the variable to be complex rather than real — and produces a theory that is strikingly more rigid and more powerful than real calculus. A function that is differentiable once on an open set is automatically differentiable infinitely often and equal to its own Taylor series; the value of such a function on a closed contour determines it everywhere inside; and integrals that are difficult or impossible to evaluate by real methods reduce to a finite sum of residues.

The course develops this theory from the ground up and then puts it to work. Topics are treated in the following order:

- Complex numbers: algebra, geometry, polar form, roots, and the topology of the complex plane.
- Analytic functions: limits, differentiability, the Cauchy–Riemann equations, and harmonic functions.
- Elementary functions: the complex exponential, trigonometric and hyperbolic functions, logarithms, powers, and branches.
- Complex integration: contours and contour integrals, Cauchy's theorem, and Cauchy's integral formula.
- Power and Laurent series; zeros and singularities.
- Residue theory and the calculation of real and complex integrals using residues.

The material is foundational for further study in analysis, differential equations, number theory, and mathematical physics, and the techniques — especially contour integration and conformal mapping — are used routinely in engineering and applied science.

Course learning outcomes

By the end of this course, a successful student will be able to:

- **LO 1.** Work fluently with complex numbers and the elementary complex functions, including the exponential, logarithmic, trigonometric and power functions, and identify their branches and domains of analyticity.
- **LO 2.** Determine whether a given function is analytic or harmonic using the Cauchy–Riemann equations, and construct harmonic conjugates.

- **LO 3.** Evaluate contour integrals, both directly from the definition and by applying Cauchy's theorem and Cauchy's integral formula.
- **LO 4.** Expand functions in power and Laurent series, and use these expansions to locate and classify zeros and singularities.
- **LO 5.** Calculate residues and apply the residue theorem to evaluate contour integrals and definite real integrals.

Essential requirements

All five learning outcomes are essential requirements of this course: each is core mathematical content that a student must demonstrate in order to pass, with or without accommodations. The standards themselves cannot be waived, but the way a student demonstrates them can be adapted through an approved accommodation — for example through extended time, alternative formats, or a different testing environment.

4. Assessment and Evaluation

Assessment summary

Assessment	Date	Weighting	CLOs assessed
In-class quizzes (12, best 8 count)	Unannounced, during lectures and tutorials throughout the term	10% of final grade	LO 1 – LO 5
Midterm Test 1	6 October, 18 – 19:30	20% of final grade	LO 1, LO 2
Midterm Test 2	18 November, 18 – 19:30	20% of final grade	LO 3, LO 4, LO 5
Final examination	TBA — scheduled by the Office of the University Registrar within the examination period, 10 – 23 December 2026	50% of final grade	LO 1 – LO 5

Note on dates. The final examination is scheduled centrally by the Office of the University Registrar; students should not make travel arrangements until the examination timetable is published.

In-class quizzes (10%)

There will be 12 pop-up quizzes at the end of lectures and tutorials. The quizzes will not be announced ahead of time and will serve in part as an attendance check. Each quiz consists of a single multiple-choice question on material covered in that lecture or the immediately preceding ones. Quizzes will be administered on OnQ.

The lowest four quiz scores are dropped, so only the best 8 of 12 count. This dropped-score provision is the accommodation for short-term extenuating and scheduling conflicts. No further accommodation, make-up quizzes, or reweighting will be provided for missed quizzes, and no documentation is required or accepted.

Midterm tests (20% each)

There will be two in-person, proctored midterm tests of 75 minutes each. No make-up midterms will be offered. A student who misses a midterm for a legitimate reason, and who has an approved academic consideration or accommodation in place (see Section 8), will have that test's weight redistributed to the remaining components — the quizzes, the other midterm, and the final examination — in proportion to their existing weights. A student who misses a midterm without an approved consideration will receive a grade of zero on that test.

Final examination (50%)

The final examination is an in-person, proctored, 180-minute examination covering the whole course. It is scheduled by the Office of the University Registrar during the December examination period. Deferred examinations are granted only through the formal academic consideration process of the student's home faculty.

Late and missed work

There is no submitted homework in this course, so no late-submission penalties apply. Suggested exercises and posted problem sets are for practice and are not collected or graded. Policies for missed quizzes, midterms, and the final examination are stated in the three sections immediately above.

Alignment of assessments with learning outcomes

The quizzes check, week by week, that the definitions and basic techniques of the current topic have been absorbed; they are formative and lightly weighted for exactly that reason. Midterm Test 1 concentrates on the foundations — complex numbers, analyticity, elementary functions, and the beginnings of contour integration (LO 1 – LO 2). Midterm Test 2 concentrates on series and singularities and the residue calculus that follows from them (LO 3 – LO 5). The final examination is cumulative and covers all material (LO 1 – LO 5).

Feedback

Midterm tests will be marked and returned as soon as possible, with a full solution set posted on onQ. Students are welcome to bring marked work to office hours to discuss it. Requests for regrading must be made in writing within one week of the work being returned and must identify the specific question and the specific concern.

5. Course Schedule

The schedule will be posted as the class progresses. At the end of each week there may be slight adjustments reflecting the actual material covered in class.

6. Required Materials and Costs

There are no required materials for this course, and no required cost to students. No lecture notes will be posted and problem sets and solutions will be posted on onQ at no charge.

Recommended textbook: sets and solutions E.B. Saff and A.D. Snyder, Fundamentals of complex analysis, 3rd edition, Pearson Education Inc., 2003. Price: 199.99 printed or 71.99 for a 180-day e-book rental.

The textbook is on reserve at Stauffer Library, and several open-access complex analysis texts of comparable coverage are available for students who prefer not to purchase a book.

Field trips

There are no field trips in this course, and no associated costs.

7. Use of Generative Artificial Intelligence (GenAI) Tools

Position: GenAI use is permitted while studying, but not permitted on any assessment.

What is permitted

You may use generative AI tools, along with any other technology, while you are learning the material on your own time: on the suggested exercises, on the posted problem sets, and while reading. This includes asking a tool to explain a definition, to work through an example, or to check an algebraic step. No disclosure or citation is required for this kind of use, because none of it is submitted for grades.

What is not permitted

No aids of any kind — no generative AI tools, no calculators, no notes, no devices — may be used during the in-class quizzes, the midterm tests, or the final examination. These assessments account for 100% of the course grade. Using a prohibited aid during an assessment, or submitting AI-generated content as your own work, is a departure from academic integrity as defined by the University's academic integrity procedures and will be handled accordingly.

A word of caution

I want to be direct: I discourage you from relying on generative AI in this course, even where it is permitted. There are two reasons. The first is that these tools are unreliable on exactly the material we are covering. They routinely produce confident, fluent, wrong complex analysis — mishandled branch cuts, invented residues, contours that do not close, theorems applied where their hypotheses fail. If you cannot yet tell a correct argument from a plausible-sounding one, you cannot tell when the tool has misled you, and that is precisely the skill this course is meant to build.

The second reason matters more. Mathematical understanding is built by struggling with a problem you cannot immediately solve, and every time you hand that struggle to a machine you skip the part where the learning actually happens. A worked solution you have read is not the same as a problem you have solved, and the difference becomes visible in a 75-minute test where no aids are allowed. Students who outsource the exercises tend to discover this in October, at some cost. Read the textbook, work the problems, get stuck, and come to office hours — that route is slower and it works.

8. University Policies and Resources

Standard institutional policies and resources — including academic integrity, accessibility and academic accommodation, technology use, and mental health and wellness — are available on the Queen's [Common Course Policies and Statements](https://www.queensu.ca/provost/teaching-and-learning/course-policies) webpage: <https://www.queensu.ca/provost/teaching-and-learning/course-policies>. These policies apply to this course and form part of this outline. The summaries below highlight the ones that come up most often.

Academic integrity

Queen's students, faculty, administrators and staff all have responsibilities for upholding the fundamental values of academic integrity: honesty, trust, fairness, respect, responsibility and courage. These values are central to the building, nurturing and sustaining of an academic community in which all members will thrive. Adherence to the values expressed through academic integrity forms a foundation for the "freedom of inquiry and exchange of ideas" essential to the intellectual life of the University (see the Senate Report on Principles and Priorities).

Students are responsible for familiarizing themselves with the regulations concerning academic integrity and for ensuring that their work and their behaviour conform to the principles of academic integrity. Departures from academic integrity include plagiarism, use of unauthorized materials, facilitation, forgery and falsification. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assessment, to failure of the course, to a requirement to withdraw from the University.

Students registered in Arts and Science should consult Academic Regulation 1 in the Arts and Science Calendar; students registered in Smith Engineering should consult Academic Regulation 1 in the Faculty of Engineering and Applied Science Calendar.

Academic consideration for students in extenuating circumstances

Queen's University is committed to providing academic consideration to students experiencing extenuating circumstances that are beyond their control and are interfering with their ability to complete academic requirements related to a course for a short period of time. See the Senate Policy on Academic Consideration for Students in Extenuating Circumstances.

Each faculty administers its own process. Students registered in Smith Engineering should contact the Program Advisor (Accommodations & Considerations) at engineering.aac@queensu.ca or 613-533-6000 ext. 78013. Students registered in Arts and Science should submit a request through the Faculty of Arts and Science academic consideration portal. Please also let me know as soon as you are able, so that I can plan around it.

Accommodations for students with disabilities

Queen's University is committed to achieving full accessibility for people with disabilities. Part of this commitment includes arranging academic accommodations for students with disabilities to ensure they have an equitable opportunity to participate in all of their academic activities. The Senate Policy for Accommodations for Students with Disabilities was approved at Senate in November 2016.

If you are a student with a disability and think you may need academic accommodations, you are strongly encouraged to contact Queen's Student Accessibility Services (QSAS) and register as early as possible; important deadlines are listed on the QSAS website. Accommodations arranged through QSAS

apply to the quizzes, the midterm tests and the final examination in this course. If you encounter a barrier in this course that your accommodation does not cover — an inaccessible posted file, a problem with a room, anything at all — please contact me directly and I will address it. Course materials can be supplied in alternate formats on request.

Copyright of course materials

Course materials created by the course instructor, including all slides, presentations, handouts, tests, examinations, and other similar course materials, are the intellectual property of the instructor. It is a departure from academic integrity to distribute, publicly post, sell or otherwise disseminate an instructor's course materials or to provide an instructor's course materials to anyone else for distribution, posting, sale or other means of dissemination, without the instructor's express consent. A student who engages in such conduct may be subject to penalty for a departure from academic integrity and may also face adverse legal consequences for infringement of intellectual property rights.

Technology used in this course

This course uses onQ, the University's learning management system, for announcements, lecture notes, problem sets and solutions, and posted grades. No other platform, polling tool, or third-party software is required. Quizzes and tests are written on paper, in person; nothing in this course is submitted or proctored online. You are encouraged to use whatever technology helps you while studying, subject to Section 7; no aids of any kind are permitted during assessments.

Mental health and wellness

If you are struggling — with this course or with anything else — support is available. Student Wellness Services offers counselling, medical care, and wellness resources to all Queen's students, and Peer Support Centre and Empower Me provide additional options, including after hours. Links are on the Common Course Policies and Statements page above. Reaching out early is easier than reaching out late.

9. Expectations and Support

What I expect of you

- Attend the lectures.
- Attend the tutorials.
- Read the textbook, or the posted notes, before and after each lecture.
- Work on the suggested exercises and the posted problem sets — steadily, not in bursts before a test.
- Discuss mathematics with other students. Collaboration on practice problems is encouraged; the assessments are individual work.

What you can expect of me

- Lectures that start and end on time and stay close to the posted schedule.

- Office hours held every week as scheduled, and appointments outside them when those hours do not work for you.
- Problem sets and solutions posted on onQ throughout the term.
- Midterm tests marked and returned within a reasonable time frame.
- Email answered within two business days.

Getting help

Office hours are the single most useful resource in this course and they are almost always underused. Come with a specific question, or come with no question at all and work on problems while I am in the room. Beyond office hours: the weekly tutorial is the place to work through exercises with support; the Mathematics and Statistics Help Centre offers drop-in assistance; and Student Academic Success Services (SASS) runs free workshops on study strategies, time management and exam preparation. Talking mathematics through with other students in the course is, in my experience, the most effective study method there is.

Inclusive and respectful classroom

This classroom should be a place where every student can ask a question without hesitating. Questions are welcome at any point in a lecture, including — especially — the ones that begin "I think I missed something three steps ago." Nobody in this room knows all of this material yet; that is what the term is for. I ask that you extend to your classmates the same patience you would want extended to you, in lectures, in tutorials, and in group study. If anything in this course makes you feel unwelcome or excluded, please tell me, and I will deal with it.

Connection to future learning

Complex analysis sits underneath a great deal of what comes next. Within mathematics it is the entry point to analytic number theory, Riemann surfaces, several complex variables, and much of the modern theory of differential equations. Outside it, contour integration and conformal mapping are working tools in fluid dynamics, electrostatics, signal processing, control theory, and quantum mechanics — the Laplace and Fourier transforms you will meet elsewhere are complex analysis in disguise. The habits this course builds, of computing carefully and then asking what makes the computation work, transfer further still.

This course outline is available in accessible PDF format on onQ. Please contact the instructor if you require it in another format.